Case Report

# Giant Calcification in Soft Tissue after Shoulder Corticosteroid Injection

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***ABSTRACT.*** We describe a giant calcification near the humeral head occurring rapidly after paramethasone injection of the shoulder. Such periarticular calcifications are rarely observed and generally after triamcinolone hexacetonide injection. (*J Rheumatol 1996;***23:181–2**)

*Key Indexing Terms:*
CORTICOSTEROID INDUCED CALCIFICATIONS

Periarticular calcifications around the joints may rarely occur iatrogenically after local corticosteroid injection.

## CASE REPORT

A 28-year-old woman, 8 months pregnant, presented with mild pain of the right shoulder due to impingement syndrome with normal radiographs. She was treated by 3 local injections of paramethasone acetate in this shoulder (40 mg injection per week for 3 weeks). Her general practitioner performed these injections in the glenohumeral joint through the subacromial region. A few days after the last injection, she suddenly presented with inflammation of the right shoulder and fever of 38.5°C, suggesting septic arthritis. She was then hospitalized in our department.

On admission, the right shoulder was tender, red, and swollen with a palpable mass near the humeral head. There was no hematoma, but the shoulder was very painful during mobilization. Examination, including gynecologic examination, was normal. Laboratory investigations revealed normal white cell count (8500/mm$^3$); C reactive protein was 14 mg/l (normal < 6) and fibrinogen 4.96 g/l (normal < 4.0). Blood cultures remained sterile. No fluid was aspirated during the glenohumeral joint aspiration. There were no biological signs of metabolic disease (calcemia was 2.40 mmol/l, phosphoremia 1.32 mmol/l, albumin 42 g/l, and protein level 71 g/l). Radiographs of the right shoulder showed a soft tissue mass near the humeral head. Glenohumeral joint space was normal with no erosion. These radiographs were performed 4 weeks after the first. Ultrasonography revealed the same mass, which was heterogeneous with microcalcifications. Bone scintigraphy was not possible due to the pregnancy. Her fever decreased rapidly and spontaneously, and she was treated only with paracetamol.

She was seen again 2 months later, after the childbirth. She still complained of right shoulder pain but it was mild and without fever. A new radiograph showed a giant calcification in the anterolateral part of the shoulder soft tissue (Figure 1). Computed tomography also showed this massive calcification (Figure 2). Laboratory inflammation tests became normal. She was seen again 5 months after the initial corticosteroid injection and the calcification remained visible on radiographs.

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## DISCUSSION

Occasionally intraarticular corticosteroid injection can induce arthritis due to crystal inflammation. But periarticular calcification following such injection is a rare complication. This has been well described with long acting corticosteroid, such as triamcinolone hexacetonide, and more rarely with other sustained action corticosteroids.

The frequency of such iatrogenic calcifications is not known precisely; a few studies have evaluated it in 26 to 50% of treated patients[1-4].

The underlying diseases requiring such articular injection are rheumatoid arthritis[1,2,4,5], juvenile rheumatoid arthritis[3,5], osteoarthritis[6,7], or spinal synovial cyst[8,9]. These calcifications concern principally the small joints, such as distal or proximal interphalangeal joints of the hand or foot. This

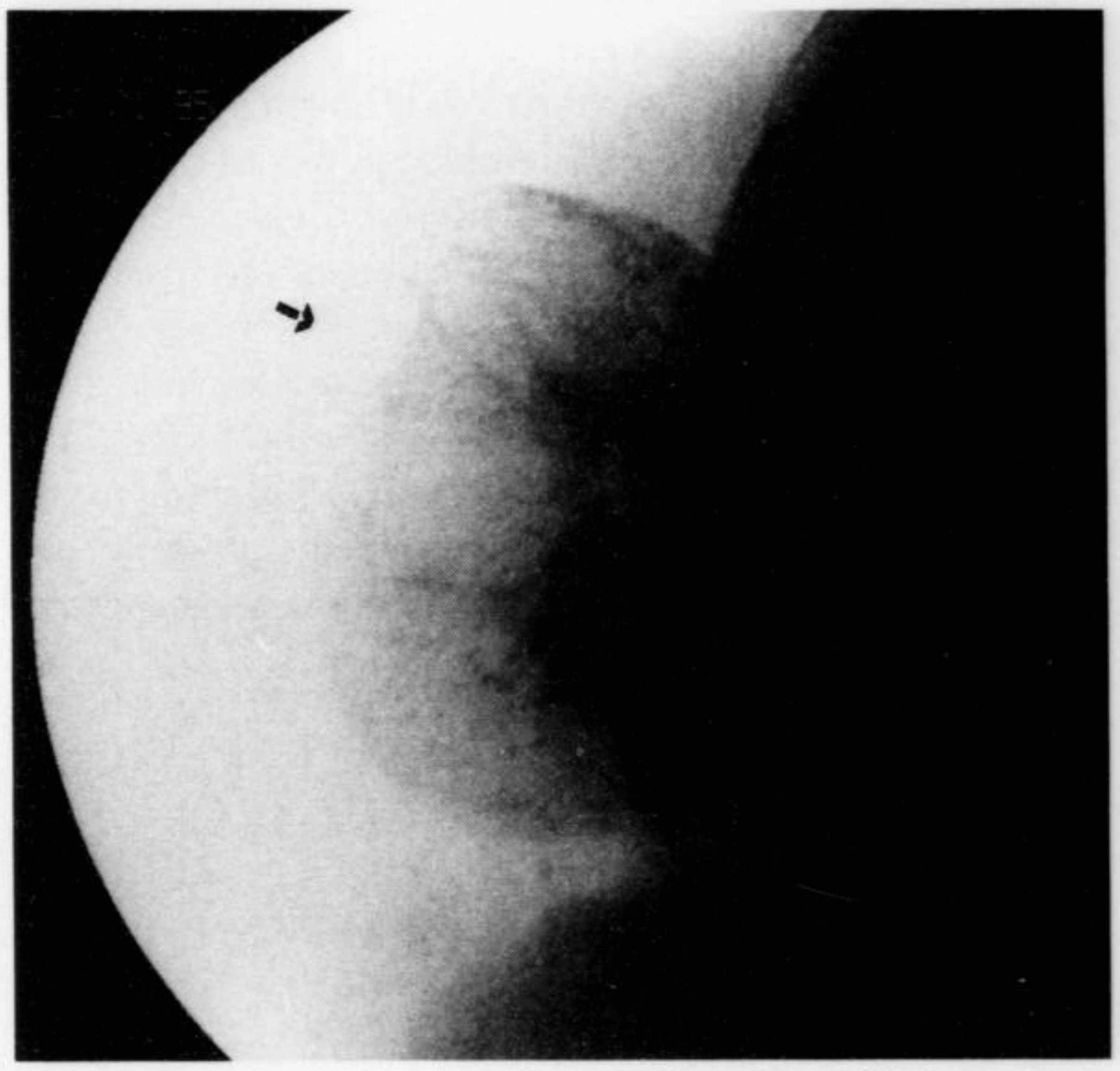

*Figure 1.* Anteroposterior right shoulder radiograph: calcified mass located in the anterolateral soft tissue of the shoulder.

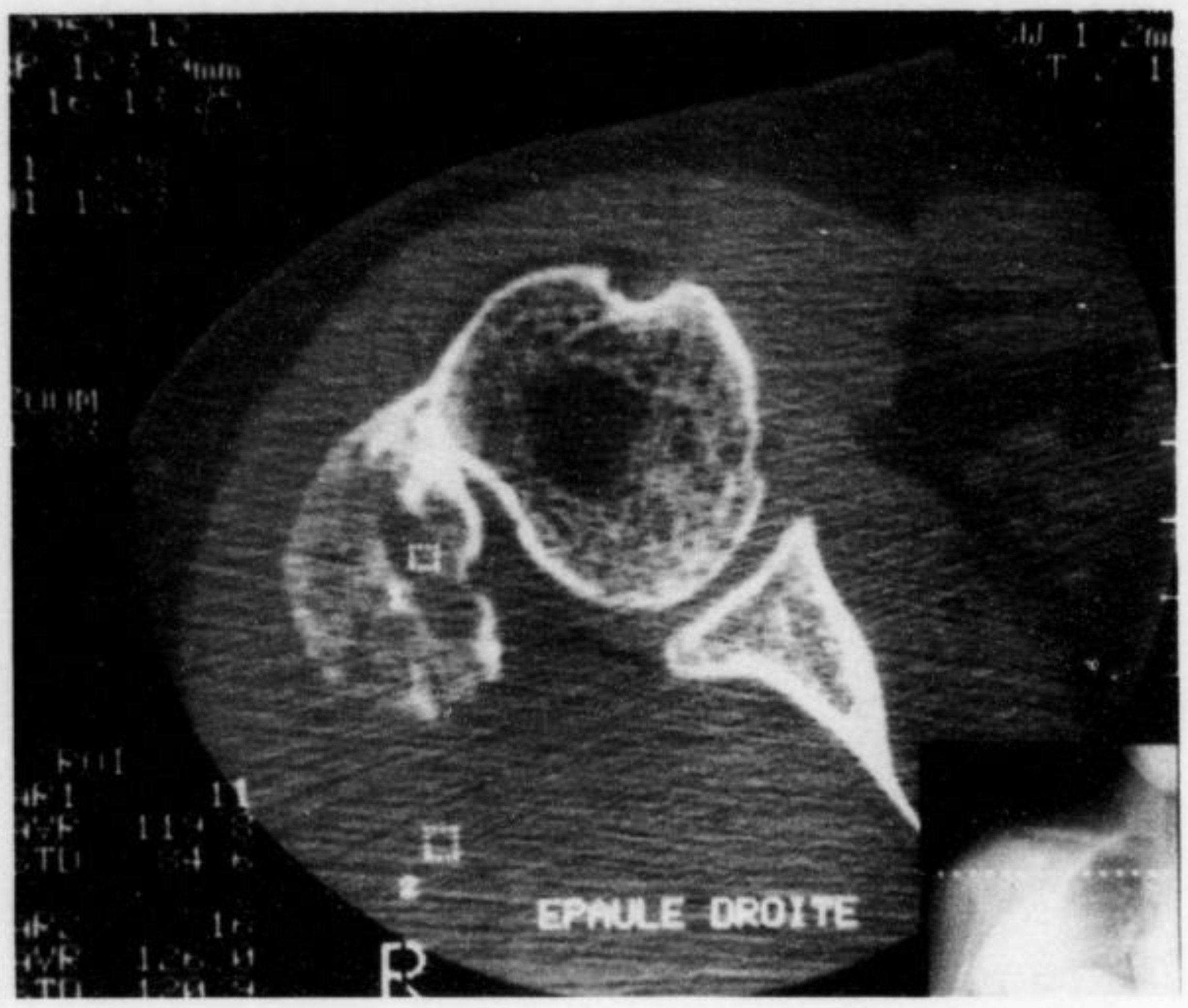


*Figure 2.* Computerized tomographic scan of the shoulder (plane in the middle of the humeral head): giant calcification adjacent to the humeral head in the soft tissue.

complication occurs generally some months after injection (2 months to 3 years).

The calcifications are revealed on radiographs performed for persistent and mild articular pain. However, they can also be asymptomatic[2] or, conversely, responsible for joint destruction[5,6]. They are located around the joint, in the capsule, tendons, ligaments, and subcutaneous tissue and vary in size and shape. Initially, they are fine and tend to become more voluminous. They can also diminish but generally remain stable[2]. However, there is rarely comment in reports on the followup or resorption of the calcification. Ultrastructural analysis of these calcifications has revealed hydroxyapatite crystals[2,6].

The pathogenesis of these periarticular calcifications is supposed to result in a dystrophic mechanism[2,3,5]. In soft tissue, corticosteroids with high atrophic power can induce an inflammatory reaction leading to necrosis, and can facilitate crystal deposits. Histologic examination of soft tissue has revealed inflammation, necrosis, and calcifications. Of interest, the contiguous synovium does not present signs of inflammation[2,3]. Metabolic disease has not been associated with these calcifications in case reports. Furthermore, the calcifications are mainly observed at the site of needle perforation, emphasizing that the drug causes local damage and leads to necrosis and calcification. Thus, long acting corticosteroids in soft tissue probably induce a nucleating site for the deposition of apatite crystals[10].

Our case report is unusual as the time between the last injection and the appearance of calcifications was very short (4 days). We hypothesize in our patient a mechanism predisposed to calciphylaxis because a hematoma has probably been induced during soft tissue injection and then precipitated crystal deposits.

To our knowledge this is the first report of calcifications occurring after paramethasone injection. Iatrogenic calcifications in soft tissue have been well described with triamcinolone hexacetonide and with betamethasone dipropionate[6]. The very low calcification frequency with these 2 corticosteroids is probably related to the short time of action (8 to 10 days, respectively) compared to triamcinolone hexacetonide (20 to 60 days)[11].

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